

Zoho round 3

Problem Statement

We are given a linear set of **8 discrete positions (dots)** numbered from 0 to 8.

Two balls are placed on these positions:

- **Cue Ball** – initially at position 0
- **Target Ball** – initially at position 2

The user provides an integer input k , which represents the number of positions the **cue ball attempts to move forward**.

The objective is to **push the target ball to position 8 or beyond** by following specific movement rules.

Movement Rules

1. The **cue ball always moves first**.
2. The cue ball attempts to move forward by k positions.
3. If the **target ball lies in the path** of the cue ball's movement:
 - The cue ball stops **one position before** the target ball.
 - The **remaining movement steps** are transferred to the target ball.
4. If the target ball is **not in the path**, the cue ball completes its movement normally.
5. The game is considered successful when the **target ball reaches position ≥ 8** .

Test case 1:

Initial position:

0.2.....

0: Cue ball

2: target ball

Input: 5

. 0 2 . .

Cue at 0

Target at 2

Gap = 2

Cue can move only **1 step** (to position 1)

Remaining steps = $5 - 1 = 4$

Target moves **4 positions** $\rightarrow 2 + 4 = 6$

Input: 3

. . . . 0 . 2 . .

Cue at 1

Target at 6

Gap = 5

No collision ($3 < 5$)

Cue moves normally

Input: 4

. 0 . . . 2

Cue at 4

Target at 6

Gap = 2

Cue moves 1 step $\rightarrow$ stops at 5

Remaining = $4 - 1 = 3$

Target moves 3 $\rightarrow 6 + 3 = 9$

Target reached

Test case 2:

Initial position:

0 . 2

Input: 2

.. 0 . 2

Input: 3

... 0 .. 2 ..

Input: 5

..... 0 2

Target reached

Implement using arrays.